

Machine Learning

Probability Basics - Week 2

Objectives

- Probability Basics
- Some exercises
- Reading

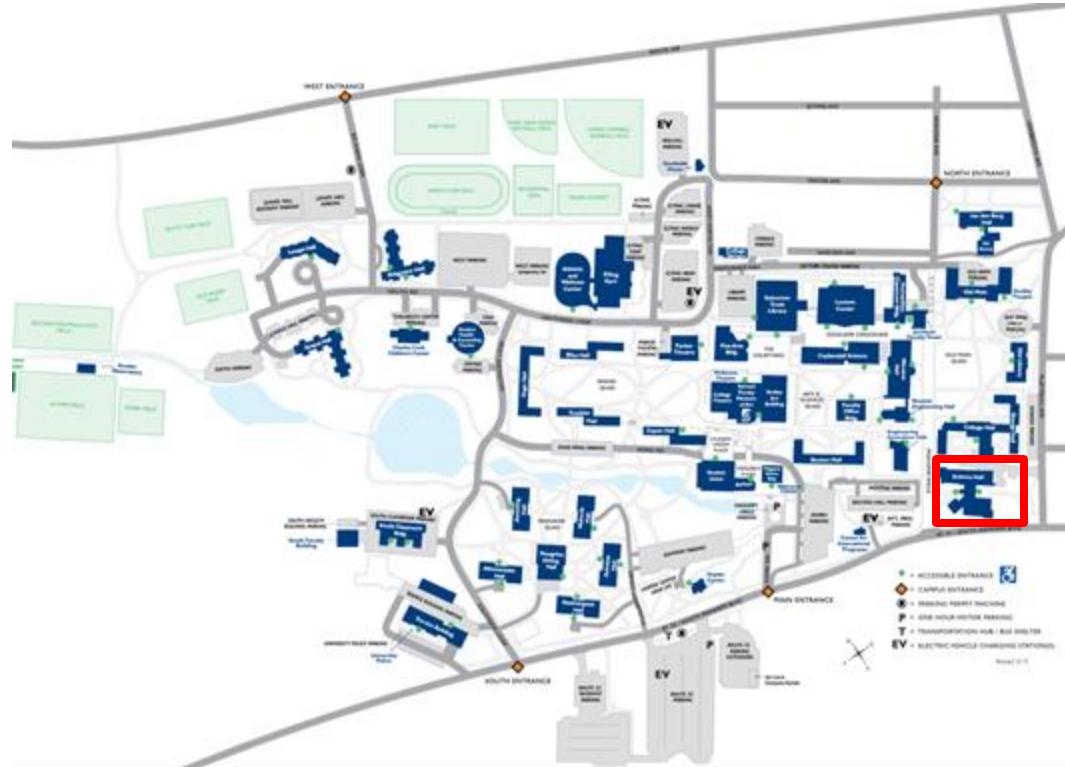
Where to find me?

Office:

Science Hall 250

Office Hours:

**Weds and Thurs 11:00 AM -
1:00 PM EST or by
appointment**



Let's look at an example

Let's imagine you are flipping a **fair** coin. There are two possible outcomes – head (H) or tail (T)

1. How likely are you going to get a head? → Event, H
2. How likely are you going to get the tail? → Event, T

In the most simple way, probability is a measure to answer these questions.

Probability is a branch of [mathematics](#) and [statistics](#) concerning [events](#) and numerical descriptions of how likely they are to occur. The probability of an event is a number between 0 and 1; the larger the probability, the more likely an event is to occur. [Wikipedia]

Some Terminology and Notation

*Let's imagine you are flipping a **fair** coin. There are two possible outcomes – head (H) or tail (T).*

We often write the total possible outcomes as $\{T, H\}$
or $\{H, T\}$ (the order doesn't matter here.) $\rightarrow$ sample set of coin tossing experiment.

$P(H)$ means Probability of throwing a head, $P(T)$ means probability of throwing a tail.

Note that $P(H) + P(T)$ is always 1. In general, for any events the probability always adds up to 1.

Also note, $P(H) = 1 - P(T)$ i.e. $P(H)$ is a complement of $P(T)$.

For a **fair** coin $P(H) = P(T) = 0.5$

Examples

Now let's toss a fair coin twice,

1. What would be the sample space here? → {TT, TH, HH, HT}
2. What are the chances of getting head twice? →
3. What is the probability of getting at least one head? →

In general, we can calculate probability of an event A as:

$$P(A) = \frac{\text{Number of favorable outcomes to A}}{\text{Total number of outcomes}}$$

Exercises

1. Flip a fair coin thrice. What is probability of getting exactly one H?
2. Throw a fair dice. What is $P(\text{Even roll})$? (this means the dice comes up with an even number on top).
3. Throw fair dice twice. What is $P(\text{a Double})$? (this means both throws resulted in the same number on top)
4. Suppose you have a biased coin with: $P(H)=.6$, $P(T)=.4$. You flip the coin twice :
 1. What is the probability of getting two H?
 2. What is the probability of at least one T?



Answers

Flip a fair coin thrice. What is probability of getting exactly one H?

Sample space: {TTT, TTH, THT, THH, HTT, HHT, HTH, HHH}.

You can also create a table

Throw 1	Throw 2	Throw 3
T	T	T
T	T	H
T	H	T
T	H	H
H	T	T
H	T	H
H	H	T
H	H	H

Event: exactly one H



$P(\text{exactly one head}) = 3/8$

Answers

Tossing a fair dice once. What is $P(\text{Even roll})$? (this means the die comes up with an even number on top).

Sample Space = $\{1,2,3,4,5,6\}$

Event, even roll = $\{2,4,6\}$

$P(\text{Even roll}) = 3/6$



Answers

Throw fair dice twice. What is $P(\text{a Double})$? (this means both throws resulted in the same number on top)

Each tossing of the dice has 6 outcomes.

Total number of outcomes = $6 * 6 = 36$. We can visualize this in a table:

		Toss 1					
		1	2	3	4	5	6
Toss 2	1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
	2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
	3	(3,1)		(3,3)			
	4				(4,4)		
	5					(5,5)	
	6						(6,6)

Now, $P(\text{a double}) = 6/36 = 1/6$

****All double events are bolded. Please note I didn't fill up the table entirely, only some cells are filled to show as an example.**



Important Point to Note

The total number of outcomes is calculated using the multiplication principle. This principle states that if there are n ways to do one thing and m ways to do another, then there are $n * m$ ways of doing both.

This can be extended to any number of events. To find the total number of outcomes, you simply multiply the number of possibilities for each individual event.

Answers

Suppose you have a biased coin with: $P(H)=.6$, $P(T)=.4$. You flip it twice:

1. What is the probability of getting two H?
2. What is the probability of at least one T?

Since, outcomes of tossing the coins are **independent**,

$$P(HH) = P(H) * P(H) = 0.6 * 0.6 = 0.36$$

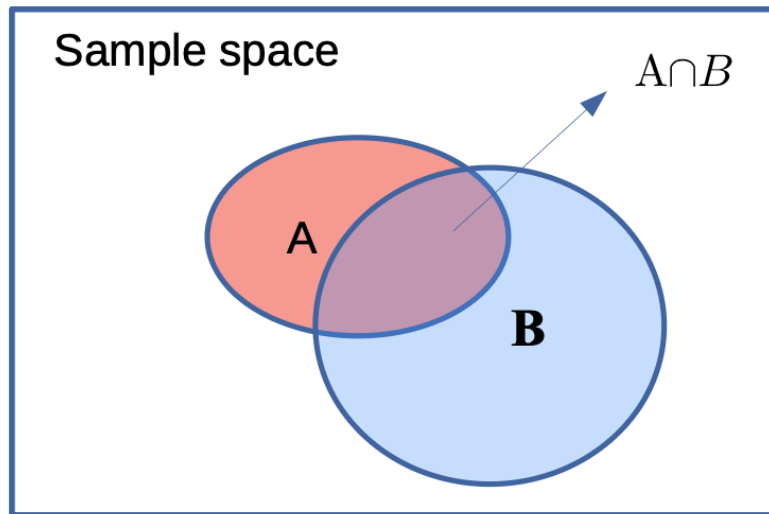
$$P(\text{at least one T}) = 1 - P(HH) = 1 - 0.36 = 0.64$$

See, the following video for reference:

https://www.youtube.com/watch?v=-FTZs_1skU

Some more terminology

- Probabilistic events are best visualized with Venn diagrams (from set theory).
- The set of all possible outcomes of an experiment (or event) is called a **sample space** of the experiment.
- An event is a subset of the sample space. An event is one of the possible outcomes of an experiment.

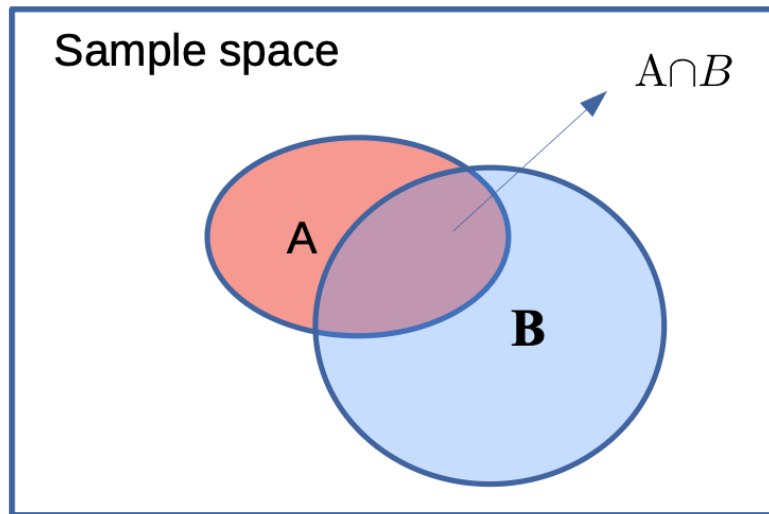


****The rectangle is a whole sample space. A and B are just two events**

Some more terminology

- ▶ The union of the two events is A or B, $A \cup B$
- The intersection of the two events, $A \cap B$
- Union is an **OR** operation; intersection is an **AND** operation

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

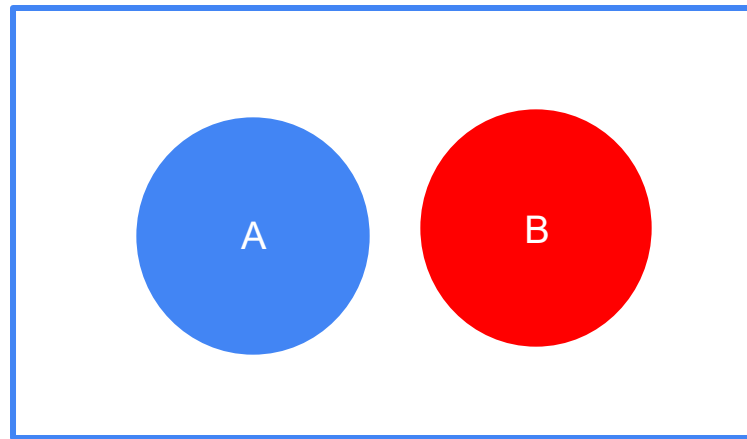


****The rectangle is a whole sample space. A and B are just two events**

Mutually Exclusive Events

- ▶ The events cannot occur together. i.e., there is no intersection between the events. E.g. Flipping a coin.
- ▶ For mutually exclusive events,

$$P(A \cup B) = P(A) + P(B) \text{ since } P(A \cap B) = 0$$



*****The rectangle is a whole sample space. A and B are just two events***

Independent Events

- ▶ The occurrence of one event doesn't affect the occurrence of the other event.
- ▶ We cannot visualize independent events through Venn diagram!
- ▶ For independent events, $P(A \cap B) = P(A) * P(B)$

Conditional Events

- ▶ In situations where we have some partial information of an outcome of an experiment, but not the whole information, we have what is called a **conditional** event.

Suppose we know that the dice toss resulted in an even number (event A). We may be interested in the probability of the toss being greater than 2 (event B). We use the notation:

$$P(B|A) \rightarrow \text{"Probability of B given A"} = P(A \cap B) / P(A)$$

Since A already, the effective set of outcomes, or sample space, has now shrunk to {2, 4, 6}. Of these, only 4 and 6 are greater than 2, resulting in $P(B|A) = 2/3$

Conditional Events

- ▶ If A and B are independent, then $P(B|A) = P(B)$ and $P(A|B) = P(A)$
- ▶ If A and B are mutually exclusive, $P(B|A) = P(A|B) = 0$

The concept of total probability

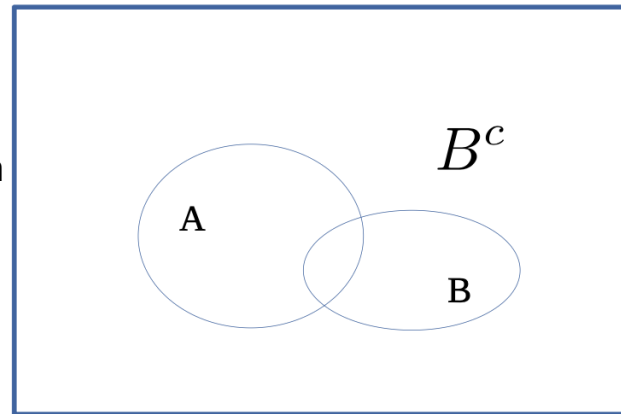
It is often useful to express the probability of some event, A , in terms of the probability of some other event, B . By looking at the Venn diagram, we can see how to express $P(A)$ in terms of conditionals:

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

Since $P(A \cap B) = P(A|B) * P(B)$ and $P(A \cap B^c) = P(A|B^c) * P(B^c)$, we can write:

$$P(A) = P(A|B) * P(B) + P(A|B^c) * P(B^c)$$

This expression is called **law of total probability**.



ANY QUESTIONS?